

Title: YouTube Trending Video Analysis Using Python

1 Introduction

YouTube is one of the world's largest video-sharing platforms where millions of users watch and upload videos every day. Some videos become trending because of high views, likes, shares, and audience engagement.

This project focuses on analyzing YouTube trending videos using Python to understand viewer behavior, popular categories, and factors that influence video trends.

2 Objectives

The main objectives of this project are:

- To analyze trending YouTube videos using dataset records
 - To identify top viewed videos and channels
 - To study relationship between likes, views, and comments
 - To determine which categories are most popular
 - To generate meaningful business insights through data analysis
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3 Dataset Description

The dataset used for this project is **USvideos.csv** downloaded from Kaggle.

Dataset Contains:

- Video ID
 - Title
 - Channel Title
 - Category ID
 - Publish Time
 - Tags
 - Views
 - Likes
 - Dislikes
 - Comment Count
 - Thumbnail Link
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Tools & Technologies Used

- Python
 - Google Colab
 - Pandas
 - Matplotlib
 - Seaborn
 - Kaggle Dataset
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Methodology

The project was completed in the following steps:

Step 1: Data Collection

Downloaded YouTube trending dataset from Kaggle.

Step 2: Data Upload

Uploaded dataset into Google Colab.

Step 3: Data Cleaning

- Removed null values
- Removed duplicate records
- Checked column names and data types

Step 4: Data Analysis

Performed analysis such as:

- Top 10 most viewed videos
- Most popular channels
- Likes vs views comparison
- Comment trends

Step 5: Visualization

Created charts and graphs for better understanding.

Exploratory Data Analysis

Top Viewed Videos

The highest viewed videos were identified using sorting techniques.

Popular Channels

Channels with most trending videos were found using count analysis.

📌 Likes vs Views

Scatter plot showed a positive relationship between likes and views.

📌 Comments Analysis

Highly engaged videos had larger comment counts.

7 Visualizations Created

- Bar Chart: Top 10 Most Viewed Videos
 - Bar Chart: Top Trending Channels
 - Scatter Plot: Likes vs Views
 - Histogram: Views Distribution
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8 Key Findings

- Entertainment and music videos dominate trending lists.
 - Popular channels trend multiple times.
 - Videos with more likes usually receive more views.
 - Viewer engagement plays an important role in trending performance.
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9 Business Insights

- Content creators should focus on engaging thumbnails and titles.
 - Upload consistency can improve channel visibility.
 - High audience interaction boosts video ranking.
 - Trending analysis helps marketers understand viewer interests.
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10 Conclusion

This project successfully analyzed YouTube trending videos using Python and Google Colab. It provided useful insights into viewer behavior, popular content types, and engagement metrics. The analysis demonstrated how data can be used to understand trends and support better content strategies.

11 Future Scope

- Use machine learning to predict trending videos

- Analyze sentiment of video titles/comments
 - Compare multiple countries' trends
 - Build interactive dashboard using Power BI/Tableau
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1 2 References

- Kaggle YouTube Trending Dataset
- Python Documentation
- Pandas Library
- Matplotlib Library